



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For



Automate Video and Voice AI Agents with n8n

TGS Ref No: TGS-2024052081

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version v4.0

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
v1.0	8 July 2026	Initial release.	Dr Alfred Ang
v2.0	11 July 2026	Agentic AI loop engineering rewrite; labs 1-6.	Dr Alfred Ang
v3.0	12 July 2026	Aligned 100% to the runnable labs: RAG concepts, Retell + ngrok setup, knowledge base and voice cloning, Vapi FAQ agent, Wav2Lip/MuseTalk/HeyGen lip-sync comparison, interactive avatars (in-browser and LiveAvatar), and Gemini Veo 3 video generation.	Dr Alfred Ang
v4.0	14 July 2026	Rebuilt as three topics over two days, aligned to the WSQ skill Artificial Intelligence Application (AER-TEM-4026-1.1): Topic 1 Chatbot (Labs 0-3), Topic 2 Voice Agent (Labs 4-5, ElevenLabs + Vapi), Topic 3 Video Agent (Labs 6-10). Added learner introductions, WSQ learning outcomes and the GitHub lab download page; matched the v4.0 WA/CS assessment set.	Dr Alfred Ang

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Course profile

Provider	Tertiary Infotech Academy Pte Ltd
Course code	TGS-2024052081
WSQ skill	Artificial Intelligence Application (AER-TEM-4026-1.1)
Version	v4.0 (14 July 2026)
Mode	Instructor-led adult training, hands-on labs
Duration	2 training days, 8 instructional hours each
Practical ratio	At least 70 percent hands-on lab time
Trainer	Dr Alfred Ang

Trainer strategy

Demonstrate each concept with one small working example, then move quickly into learner practice. For every lab, ask learners to show the evidence - the n8n execution trace, the generated media, the call transcript, the scorecard. Do not accept a screenshot of a finished screen as proof that the workflow is correct: a fluent output and a correct workflow are not the same thing, and telling them apart is the skill this course teaches.

Topic map

Each topic delivers one WSQ learning outcome of Artificial Intelligence Application (AER-TEM-4026-1.1). The Written Assessment (SAQ) tests K1-K6; the Case Study tests A1-A6.

Session	Topic	Focus	WSQ outcome	Labs
1	Topic 01 - Chatbot	Agents, retrieval and grounded answers: your first n8n AI agent, a RAG chatbot that admits what it	LO1 (K1, K6, A3, A5)	Lab 0, Lab 1, Lab 2, Lab 3

		does not know, and a customer-facing course advisor on a real website. Labs 0-3.		
2	Topic 02 - Voice Agent	Two vendors, two architectures: ElevenLabs runs the model and calls your n8n tools; with Vapi your n8n workflow IS the model. The contrast is the lesson. Labs 4-5.	LO2 (K2, K3, A1, A4)	Lab 4, Lab 5
3	Topic 03 - Video Agent	Lip-sync, avatars and text-to-video: from a script, to a mouth that moves, to a face that talks back, to video from nothing but a sentence. Labs 6-10.	LO3 (K4, K5, A2, A6)	Lab 6, Lab 7, Lab 7-os, Lab 8, Lab 9, Lab 10

Daily schedule

Total taught content: 12 hours 00 minutes across 12 labs.

Day 1

Time	Duration	Topic / Activity	Deck	Evidence produced
9:30am - 10:15am	45 min	Lab 0 - Set Up n8n Locally	Slide 23	A running local stack: screenshots of

				docker compose ps, ollama list and the passing credential test.
10:15am - 11:00am	45 min	Lab 1 - Your First AI Agent	Slide 29	A working local agent plus a one-line note on how the system prompt changed its behaviour.
11:00am - 12:00pm	60 min	Lab 2 - RAG IT Support Chatbot	Slide 35	A RAG chatbot answering from the IT FAQ, plus one provoked refusal captured in the execution trace.
12:00pm - 1:00pm	60 min	Lab 3 - CX Agent with RAG (Cook & Bake Academy)	Slide 42	A working course-advisory chatbot on the site, with one grounded answer and one honest refusal in the trace.
1:00pm - 2:00pm	60 min	LUNCH	-	-
2:00pm - 4:00pm	120 min	Lab 4 - Voice Booking Agent with ElevenLabs (GG Hair Salon)	Slide 55	A real calendar booking made by voice: the calendar event, the call transcript, and the n8n tool execution.
4:00pm - 5:30pm	90 min	Lab 5 - Grounded FAQ Voice Agent with Vapi (MediRefill)	Slide 62	A live Vapi call transcript showing grounded answers, a hard medical refusal, and the emergency escalation.

5:30pm - 6:30pm	60 min	Concept delivery, review and evidence check - the trainer teaches the concept, workflow and quality slides for the day's labs, then learners show their executions, transcripts and scorecards	-	Trainer sign-off on the day's evidence
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Day 1 instructional time: 8h 00m (9:30am-6:30pm, less a 1-hour lunch; tea breaks are taken inside the sessions).

Day 2

Time	Duration	Topic / Activity	Deck	Evidence produced
9:30am - 10:30am	60 min	Lab 6 - Lip-Sync Face-Off: MuseTalk vs HeyGen	Slide 73	Two rendered clips of one script, a scored comparison, and a one-line recommendation per engine.
10:30am - 11:15am	45 min	Lab 7 - Avatar News Video with HeyGen (GG News Studio)	Slide 80	A generated avatar news video plus quality-review notes on the script and the render.
11:15am - 12:00pm	45 min	Lab 7-os - Open-Source News Avatar - Free and Local	Slide 86	A locally rendered news video and a written HeyGen-versus-local comparison.
12:00pm -	45 min	Lab 8 -	Slide 92	A conversation

12:45pm		Interactive Avatar Brain (Aria, In-Browser)		with Aria, the latency-panel screenshot, and one prompt improvement you made and re-tested.
12:45pm - 1:45pm	60 min	LUNCH	-	-
1:45pm - 2:30pm	45 min	Lab 9 - Interactive Avatar Session (Nova, HeyGen LiveAvatar)	Slide 98	A working Nova session plus the two-column scorecard comparing her against the browser-rendered avatar.
2:30pm - 3:30pm	60 min	Lab 10 - AI Video Generation with Gemini Veo 3 (Veo Studio)	Slide 104	A rendered Veo clip, the shot prompt that produced it, and a note on what changed between two prompt versions.
3:30pm - 4:15pm	45 min	Concept delivery, review and evidence check - the trainer teaches the concept, workflow and quality slides for the day's labs, then learners show their executions, transcripts and scorecards	-	Trainer sign-off on the day's evidence
4:15pm - 4:30pm	15 min	Assessment briefing - instruments, timing, and	-	Completed answer script submitted on the LMS

		what evidence is graded		
4:30pm - 5:30pm	60 min	Written Assessment (SAQ) - 6 open-ended knowledge questions (K1-K6)	-	Completed answer script submitted on the LMS
5:30pm - 6:30pm	60 min	Case Study (CS) - 6 scenario tasks drawn from the labs (A1-A6)	-	Completed answer script submitted on the LMS

Day 2 instructional time: 8h 00m (9:30am-6:30pm, less a 1-hour lunch; tea breaks are taken inside the sessions).

Assessment

Instrument	Duration	What it tests	When
Briefing for Assessment	15 min	The learner knows the instruments, the criteria and the appeal route	Before the assessment, on the final day
Written Assessment (WA) - Short Answer	60 min	KNOWLEDGE: the concepts behind the labs	Final day, after the briefing
Case Study (CS)	60 min	APPLICATION: judgement on a workplace scenario built from the labs	Final day, after the WA

Both instruments are open book, individually attempted, and marked Competent / Not Yet Competent. A learner assessed NYC is re-assessed once, on the failed instrument only.

Evidence the trainer collects

- Workflow exports and n8n execution traces for each lab.
- The provoked failures: the RAG refusal (Lab 2), the course that does not exist (Lab 3), Ava's medical refusals (Lab 5).
- The Google Calendar booking created by voice, with its call transcript (Lab 4).
- The lip-sync comparison scorecard: MuseTalk vs HeyGen on one script (Lab 6).
- Generated media: the news avatar videos (Labs 7 and 7-os), the Aria latency screenshot (Lab 8), the Nova session scorecard (Lab 9), the Veo clip and its shot prompt (Lab 10).